

ASHTON DY

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EDUCATION

- 2023 - 2027** **Bachelor of Science, Honors Computer Science** – *The University of Tennessee – Knoxville, Tickle College of Engineering*
- University Honors Program, Engineering Honors Program, Summa Cum Laude, GPA: 3.98

RESEARCH EXPERIENCE

- May 2026 - Pres.** **Research Student Intern** – *Oak Ridge National Laboratory*
- Developed and optimized visual molecular dynamics-based tools for membrane trajectory analysis.
 - Cut down the membrane trajectory process time from 3 minutes to 5 seconds.
 - Collaborated with peer interns to execute a GROMACS molecular dynamics workflow, analyzing membrane-protein systems and structural stability metrics.
- Sept. 2025 - Pres.** **Undergraduate Research Assistant** – *Dr. Coble's Lab*
- Developed a comprehensive knowledge graph to provide structured insights into sheet metal forming processes.
 - Designed and integrated knowledge graphs and vector databases into an LLM-driven RAG pipeline, leveraging Gemini and LangChain for enhanced retrieval and generation performance.
 - Curated a domain-specific sheet metal forming dataset for ingestion into both knowledge graph and vector database systems.
- May 2025 - Pres.** **Undergraduate Research Assistant** – *Dr. Vogiatzis' Lab*
- Fine-tuned Transpharmer, a transformer-based molecular generator, for de novo design of NLRP3 inhibitors relevant to Alzheimer's disease.
 - Implemented unconditional and conditional generation workflows, benchmarking against the known inhibitor GDC-2394 to evaluate scaffold novelty and clarity.
 - Integrated a custom XGBoost binding affinity predictor trained on MD simulation outputs to guide reinforcement learning.
 - Designed a reinforcement learning pipeline with policy gradient optimization, incorporating drug-likeness constraints (Lipinski, MW, logP) to prioritize realistic molecules.
- Feb 2025 - Aug 2025** **Undergraduate Research Assistant** – *AURAS Lab*
- Conducted research at the intersection of robotics and computer vision, focusing on image classification, segmentation, and motion planning.
 - Curated, labeled, and augmented 30,000+ images, boosting model training efficiency and classification accuracy.
 - Evaluated and implemented Segment Anything Model v2 (SAM2) for advanced image segmentation and key-point querying tasks.
 - Studied geometric algebra concepts and analyzed their applicability to robot motion planning.
- Mar 2024 - May 2025** **Undergraduate Research Assistant** – *Dr. Zhao's Lab & EPIC Lab*
- Conducted literature review on human–robot interaction for social and physical robots.
 - Programmed quadrupeds (Unitree Go2 & B1) using ROS.

- Designed a study on seniors' perceptions of robotic assistance; created quantitative/qualitative surveys and an interactive activity.
- Built a TensorFlow classification model after cleaning and labeling raw data.

PUBLICATIONS

2025 Azizul Zahid, Jie Fan, Farong Wang, **Ashton Dy**, Sai Swaminathan, Fei Liu. "Toward Aligning Human and Robot via Multi-Modal Demonstration Learning", *ICRA 2025 Workshop*, 2025.
<https://arxiv.org/abs/2504.11493>

PROFESSIONAL EXPERIENCE

- May 2025 - Aug 2025 **Software Engineer Intern** – *Clayton Homes*
- Spearheaded the first AI-powered internal tool, creating a Model Context Protocol (MCP)-based RAG pipeline used by 10,000+ employees.
 - Developed an MCP-powered AI chatbot with a custom web interface integrated with Confluence, SharePoint, Jira, and GitHub through AWS Kendra.
 - Deployed the backend with Docker and Elastic Container Services under an Application Load Balancer; hosted frontend via S3 + CloudFront.
 - Leveraged AWS Bedrock for cloud-hosted LLMs and implemented IAM roles, permissions, and policies.
 - Practiced Agile principles: sprint planning, ticketing, and daily stand-ups.
- Nov 2024 - Mar 2025 **Software Engineer Intern** – *Function Solutions*
- Designed and developed a HIPAA-compliant medical data tracking system for the Tennessee Department of Correction, securely managing records for 320+ patients.
 - Implemented JSON Web Tokens and data encryption to safeguard patient information and meet HIPAA standards.
- Aug 2024 - May 2025 **Resident Assistant** – *University of Tennessee Housing*
- Fostered a community of 58 upperclassmen through 13 engaging programs and 50+ individual meetings.
 - Collaborated with other staff to support residents and improve campus life.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, Java, HTML, CSS, RISC-V Assembly
Frameworks	PyTorch, TensorFlow, rdkit, pandas, MongoDB, Express.js, React.js, Node.js, SASS, Framer Motion
Tools & Cloud	Docker, AWS (ECS, S3, CloudFront, IAM, Kendra, Bedrock), Linux, Windows, GitHub, Fusion 360, Blender, Microsoft Office, Google Suite

INVOLVEMENT

- Jan. 2024 - Pres. **Events Director & Outreach Director** – *Society of Asian Scientists and Engineers*
- Designed and organized 15 events for the full academic year, collaborating with other collegiate and professional organizations.

- Mar. 2024 - Pres.

Initiation Chair – *Tau Beta Pi (Engineering Honors Society)*

 - Invitation-only engineering honors society that recognizes the top 12.5% of juniors and 20% of seniors.
- Sept. 2025 - Pres.

Active Member – *Computational Biology and Sciences Club*

 - Collaborated with others interested in computational sciences.
 - Listened to presentations from faculty and participated in workshops relating to technology in the computational sciences space.
- Sept. 2025 - Pres.

Minor Board Officer – *Lambda Phi Epsilon International Fraternity Inc.*

 - Web Chair: Developed and maintain the chapter’s website, improving accessibility and communication for members.
 - Academic Chair: Organized events promoting academic excellence, leadership development, and community engagement.
 - Multi-Cultural Council Rep: Represented the organization at Multicultural Greek Council meetings to advocate for chapter interests and foster inter-chapter collaboration.

HONORS & AWARDS

Academic	University Honors Engineering Honors (Cook Grand Challenge) Summa Cum Laude
Fall '23 - Sp '26	Dean's List
Honor Society	Phi Kappa Phi – Recognized within the Top 7.5% of Juniors